

1. What is preference based approach of consumer's behavior? Explain and interpret the ~~axioms~~ assumptions of preference based approach (e.g. Completeness, Reflexivity, Rationality)

2. Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity

3. ~~with~~ show that
strong monotonicity $\Rightarrow$ monotonicity.

4. show that

strong $\xrightarrow{\quad}$ Monotonicity $\nrightarrow$ LNS
Monotonicity $\nleftarrow{\quad}$

5. Define indifference set. Explain how does ~~the~~ strong monotonicity and indifference curve's slopes are related

6. What are the twin definitions of convexity? In what respect they are different from strong convexity. examine graphically.

7. Consider the following utility functions:

a) $u(x_1, x_2) = ax_1 + bx_2$

b) $u(x_1, x_2) = \min(ax_1, bx_2)$

c) $u(x_1, x_2) = ax_1^{1/2} \times bx_2^{1/2}$

d) $u(x_1, x_2) = ax_1^{1/2} \times bx_2^2$

Examine which function satisfies convexity, strict convexity or both.

8. Show that

Convexity of preferences $\Leftrightarrow$ $u^s c(x)$ is convex $\Leftrightarrow u(\cdot)$ is quasi-concave

9. In case of well defined utility function, show that

Concavity $\nRightarrow$ Quasi-concavity

10. What are the typical utility functions ~~is~~ used in Economic applications. Write down their functional form and calculate and interpret the notion of MRS.

11. Examine the ^{following} properties of preference relation:

i) Homogeneity

ii) Homotheticity

proof that.

12. If a preference relation satisfies monotonicity and continuity, then there exist a utility function $u(\cdot)$ representing such preference relation.

13. Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, the examine the properties satisfied by the Walrasian demand function $x(p, w)$. Examine the situation when the sufficient condition A UMP may be violated.

14. Define indirect utility function. ⁴
Examine its properties.

15. Consider the Cobb-Douglas utility function:

$$U(x_1, x_2) = K x_1^\alpha x_2^{1-\alpha}$$

for some $\alpha \in (0, 1)$, and $K > 0$

i) Derive the Walrasian demand functions and examine their properties.

ii) How do you get the indirect utility function. Explain their properties. Verify Roy's identity.

iii) Solve the expenditure maximization problem and explain the properties of the compensated demand function.

16. a) Distinguish between CV and EV.

b) How do you measure the welfare effects of a price change.

17. Graphically show that for a decrease in the price of a commodity (when both goods are normal)

$$EV > CS > CV$$

for price increase

$$EV < CS < CV.$$

18. Define production vector/~~sets~~ transformation function, and transformation frontier, production ~~sets~~.

19. Define production sets. ^{Graphically} Explain the properties of the production set.

20. ^{Define} Find the elasticity of substitution (E_s). Find the elasticity of substitution for the production function. $q = f(K, L) = aK + bL$.

What will be the Eos if the production function is fixed proportions.

21. Define profit function. If the production set Y is closed and have free disposal, then what properties do the profit function satisfy.

22. Consider a consumer with the utility function

$$u(x_1, x_2) = x_1^{1/4} x_2^{1/2}$$

i) Derive the ordinary and compensated demand functions

ii) Consider $\{p^0, m^0\} = \{4, 4, 300\}$
 $\{p', m^0\} = \{2, 4, 300\}$

a) compute the indirect utilities and costs.

ii) Calculate EV , CV (in $\text{\$}$) and examine their relationship and on the nature of the commodities.